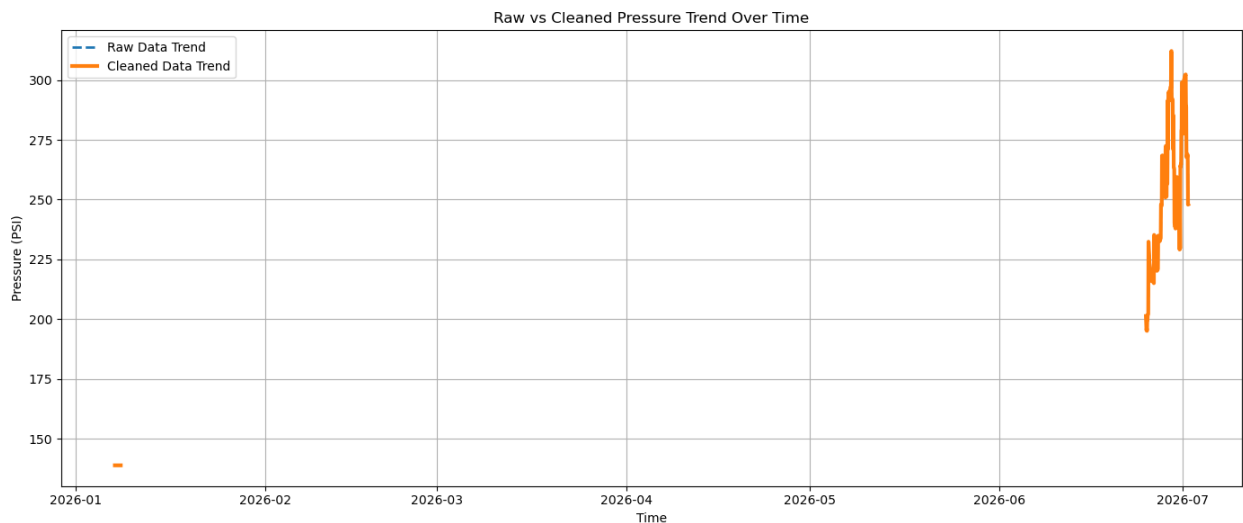


Week 2 Insight Report

Raw vs. Cleaned Sensor Data Trend Analysis



Context

The initial operational sensor dataset contained several data quality issues, including missing values, duplicate records, inconsistent categorical labels, and physically impossible sensor readings. These issues introduced noise into the analysis and reduced confidence in the reliability of reported operational performance. Without cleaning, the raw data risked producing misleading conclusions about equipment behavior and process stability.

Insight

After cleaning and standardizing the dataset, the underlying operational trend became clearer and more reliable. The cleaned data removed anomalous readings that distorted the true behavior of the system and revealed a consistent pressure pattern across operating periods. While the overall trend remained stable, the cleaning process confirmed that several apparent fluctuations in the raw data were caused by data quality issues rather than genuine operational events. This provides a more accurate view of plant performance and improves confidence in operational monitoring.

Action

I recommend implementing automated data validation checks at the point of collection to identify impossible sensor readings, duplicate records, and missing values before they enter reporting systems. In addition, supervisors should review sensor calibration schedules and investigate recurring data quality issues to prevent future anomalies. Establishing a routine data-quality monitoring process will improve operational visibility, support more accurate decision-making, and reduce the risk of responding to false alarms generated by poor-quality data.